

Supplemental Material to *Nonlinear semiclassical spectroscopy of ultrafast molecular polariton dynamics*

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S1. VECTORIZATION/MAPPING TO LIOUVILLE SPACE

It is advantageous to convert the von-Neumann (master) equation for the density matrix onto matrix-vector form, i.e., map the $n \times n$ density matrix ρ onto an n^2 vector $\vec{\rho}$ and the superoperator acting on the density matrix onto an $n^2 \times n^2$ matrix $\mathcal{L}$ (referred to as Liouville space). For instance, for a two-level system, the density matrix is mapped onto a column vector as

$$\rho = \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix} \rightarrow \vec{\rho} \equiv |\rho\rangle = \begin{pmatrix} \rho_{11} \\ \rho_{12} \\ \rho_{21} \\ \rho_{22} \end{pmatrix}. \quad (\text{S.1})$$

This can be done in a more formal way in a procedure known in the literature as vec-ing [1, 2], which has the following rules

1. A left multiplication of the matrix ρ by an $n \times n$ matrix A , i.e., $A\rho$, is equivalent to an operation on the vector $\vec{\rho}$ by the $n^2 \times n^2$ matrix $A \otimes I$, where I is the $n \times n$ identity matrix, and $\otimes$ is the Kronecker product.
2. Similarly, a right multiplication of the matrix ρ by an $n \times n$ matrix B , i.e., ρB , is equivalent to an operation on the vector $\vec{\rho}$ by the $n^2 \times n^2$ matrix $I \otimes B^\top$.
3. Finally, a combination of left and right matrices multiplication, $A\rho B$, is equivalent to an operation on the vector $\vec{\rho}$ by the $n^2 \times n^2$ matrix $A \otimes B^\top$.

With this, one can express general commutators as

$$[\mathcal{H}, \rho] \rightarrow (\mathcal{H} \otimes I - I \otimes \mathcal{H}^\top) \vec{\rho} \equiv \mathcal{L}\vec{\rho}. \quad (\text{S.2})$$

S2. CAVITY INPUT-OUTPUT RELATIONS

Let us consider a (classical) input field $\langle \hat{b}_{\text{in},L}(t) \rangle$ driving the cavity through the left mirror described by the Hamiltonian

$$\mathcal{H}_d = -i\hbar\sqrt{\kappa_L} \langle \hat{b}_{\text{in},L}(t) \rangle \hat{a}^\dagger + \text{H.c.}, \quad (\text{S.3})$$

where κ_L describes the loss rate through the left mirror. For our purposes, we consider an input pulse described by $\langle \hat{b}_{\text{in},L}(t) \rangle = \eta f(t)/\sqrt{\kappa_L}$ (in a frame rotating at the pulse frequency). The parameter η can be related to the incident laser power $\mathcal{P}_{\text{in}}$ as $\eta = \sqrt{\mathcal{P}_{\text{in}}\kappa_L/(\hbar\omega_\ell)}$, where ω_ℓ is the pulse carrier frequency. Then, the cavity input-output relations written separately at the left (L) and right (R) mirror, respectively, are given by [3]

$$\hat{b}_{\text{out},L}(t) - \hat{b}_{\text{in},L}(t) = \sqrt{\kappa_L} \hat{a}, \quad (\text{S.4a})$$

$$\hat{b}_{\text{out},R}(t) - \hat{b}_{\text{in},R}(t) = \sqrt{\kappa_R} \hat{a}, \quad (\text{S.4b})$$

from which the transmitted intensity can be straightforwardly derived as

$$T = \left| \frac{\langle \hat{b}_{\text{out},R}(t) \rangle}{\langle \hat{b}_{\text{in},L}(t) \rangle} \right|^2 = \frac{\kappa_R \kappa_L}{\eta^2 f(t)^2} |\alpha|^2. \quad (\text{S.5})$$

For a balanced cavity $\kappa_L = \kappa_R = \kappa/2$, this becomes

$$T = \frac{(\kappa/2)^2}{\eta^2 f(t)^2} |\alpha|^2, \quad (\text{S.6})$$

which holds in both linear and nonlinear regimes, as well as in the Fourier domain. It is important to note, however, that in the ultrastrong coupling regime, the input-output relations are significantly modified due to the breakdown of the rotating wave approximation (RWA). In this regime, where the collective coupling strength $g\sqrt{\mathcal{N}}$ becomes

comparable to the cavity or matter transition frequencies, non-trivial corrections to the input-output relations must be incorporated that account for counter-rotating terms and the hybrid eigenmodes of the light-matter system [4, 5].

S3. GENERAL FORM OF PUMP-PROBE EQUATIONS

For a cavity driven by two pulses, the $(n)(m)$ -th order equations for the field and molecular density matrix are given by (assuming initial conditions $\alpha(0) = \alpha^{(0)(0)} = 0$)

$$\dot{\alpha}^{(n)(m)} = -\left(\frac{\kappa}{2} + i\omega_c\right)\alpha^{(n)(m)} - iE_0\mathcal{N}\text{Tr}[\hat{\mu}\rho^{(n)(m)}] - \delta_{1n}\delta_{0m}f_p(t - \tau_p)e^{-i\omega_p t} - \delta_{0n}\delta_{1m}f_{p'}(t - \tau_{p'})e^{-i\omega_{p'} t}, \quad (\text{S.7a})$$

$$\dot{\rho}^{(n)(m)}(t) = -i[\mathcal{H}_0, \rho^{(n)(m)}] - iE_0 \sum_{j=0}^{n-1} \sum_{j'=0}^{m-1} \left(\alpha^{(n-j)(m-j')} + \alpha^{(n-j)(m-j')*} \right) [\hat{\mu}, \rho^{(j)(j')}], \quad (\text{S.7b})$$

where we will generally assume identical pulse envelopes $f_j(t - \tau_j) = e^{-(t-\tau_j)^2/\tau_w^2}$. In Liouville space, the equation for the density vector can then be expressed in compact form as

$$\dot{\vec{\rho}}^{(n)(m)}(t) = -i\mathcal{L}_0\vec{\rho}^{(n)(m)}(t) - i \sum_{j=0}^{n-1} \sum_{j'=0}^{m-1} \mathcal{L}_{\text{int}}^{(n-j)(m-j')}(t) \vec{\rho}^{(j)(j')}(t), \quad (\text{S.8})$$

where we defined the interaction Liouvillian

$$\mathcal{L}_{\text{int}}^{(n-j)(m-j')}(t) = E_0[\alpha^{(n-j)(m-j')}(t) + \text{c.c.}]\mathcal{L}_\mu, \quad (\text{S.9})$$

and the vectorized commutators are given by

$$[\mathcal{H}_0, \rho] \rightarrow \mathcal{L}_0\vec{\rho} = [\mathcal{H}_0 \otimes I - I \otimes \mathcal{H}_0^\top]\vec{\rho}, \quad (\text{S.10a})$$

$$[\hat{\mu}, \rho] \rightarrow \mathcal{L}_\mu\vec{\rho} = [\hat{\mu} \otimes I - I \otimes \hat{\mu}^\top]\vec{\rho}. \quad (\text{S.10b})$$

S4. DIFFERENTIAL TRANSMISSION

We define the differential transmission (DT) as the difference in the probe transmission with and without the presence of the pump pulse

$$\Delta T(\omega) = T_{p'}^{\text{pump on}}(\omega) - T_{p'}^{\text{pump off}}(\omega). \quad (\text{S.11})$$

Considering only the lowest (third) order correction of the probe field due to the pump, the normalized transmitted probe intensities can be expressed from input-output relations in Fourier domain (see section S2) as

$$T_{p'}^{\text{pump off}}(\omega) = \left(\frac{\kappa}{2}\right)^2 \frac{|\alpha^{(0)(1)}(\omega)|^2}{f_{p'}(\omega)^2}, \quad T_{p'}^{\text{pump on}}(\omega) \approx \left(\frac{\kappa}{2}\right)^2 \frac{|\alpha^{(0)(1)}(\omega) + \eta_p^2 \alpha^{(2)(1)}(\omega)|^2}{f_{p'}(\omega)^2}. \quad (\text{S.12})$$

From this, we can approximate the DT as

$$\Delta T(\omega) \approx \left(\frac{\kappa}{2}\right)^2 \eta_p^2 \frac{2\text{Re}[\alpha^{(0)(1)*}(\omega)\alpha^{(2)(1)}(\omega)]}{f_{p'}(\omega)^2}. \quad (\text{S.13})$$

This quantity is not the usual pump-probe signal but contains different phase contributions (see main text, paragraph ‘Differential transmission’).

S5. ANALYTICAL SOLUTION OF PUMP-PROBE DYNAMICS FOR A 2LS

For a two-level system, let us consider the (2)(1)-order equations for the cavity field and the density matrix (second order in pump, first in probe), required for the computation of the DT derived in Sec. S4 (assuming a matter initial

state $\rho(0) = |g\rangle\langle g|$

$$\dot{\alpha}^{(2)(1)}(t) = -\left(\frac{\kappa}{2} + i\omega_c\right)\alpha^{(2)(1)} - ig\mathcal{N}\rho_{21}^{(2)(1)}, \quad (\text{S.14a})$$

$$\begin{aligned} \dot{\rho}_{21}^{(2)(1)}(t) = & -\left(\frac{\gamma + \gamma_\phi}{2} + i\omega_0\right)\rho_{21}^{(2)(1)} - ig\alpha^{(2)(1)} + ig\alpha^{(0)(1)}(\rho_{22}^{(2)(0)} - \rho_{11}^{(2)(0)}) \\ & + ig\alpha^{(1)(0)}(\rho_{22}^{(1)(1)} - \rho_{11}^{(1)(1)}), \end{aligned} \quad (\text{S.14b})$$

where we considered dissipation affecting the molecules in the form of dephasing (γ_ϕ) and spontaneous emission (γ). The third-order coherence $\rho_{21}^{(2)(1)}$ is driven by the population created by the pump (2)(0) as well as by the population created by both pump and probe (1)(1). We rewrite the equation for the third-order coherence as

$$\dot{\rho}_{21,\tau}^{(2)(1)}(t) = -\left(\frac{\gamma + \gamma_\phi}{2} + i\omega_0\right)\rho_{21,\tau}^{(2)(1)} - ig\alpha_\tau^{(2)(1)} + F_\tau(t), \quad (\text{S.15})$$

where we condensed the input affecting the third-order coherence into a single term $F_\tau(t)$ and by the index τ we indicate that the quantities depend on the delay time between the pulses τ_Δ .

In Fourier space, we can therefore obtain the solution for $\alpha_\tau^{(2)(1)}(\omega)$

$$\alpha_\tau^{(2)(1)}(\omega) = \frac{-ig\mathcal{N}F_\tau(\omega)}{[i(\omega_c - \omega) + \frac{\kappa}{2}][i(\omega_0 - \omega) + \frac{\gamma + \gamma_\phi}{2}] + g^2\mathcal{N}}. \quad (\text{S.16})$$

From the equations of motion, we can relate second-order quantities to first-order quantities in frequency space as ($*$ denotes the convolution operation)

$$\rho_{22}^{(2)(0)}(\omega) - \rho_{11}^{(2)(0)}(\omega) = \frac{4ig}{\omega + i\gamma} \text{Im} \left[\left(\alpha^{(1)(0)} * \rho_{12}^{(1)(0)} \right) (\omega) \right], \quad (\text{S.17a})$$

$$\rho_{22}^{(1)(1)}(\omega) - \rho_{11}^{(1)(1)}(\omega) = \frac{4ig}{\omega + i\gamma} \left\{ \text{Im} \left[\left(\alpha^{(0)(1)} * \rho_{12}^{(1)(0)} \right) (\omega) \right] + \text{Im} \left[\left(\alpha^{(1)(0)} * \rho_{12}^{(0)(1)} \right) (\omega) \right] \right\}, \quad (\text{S.17b})$$

and therefore, we can express $F_\tau(\omega)$ as a sum of double convolutions

$$\begin{aligned} F_\tau(\omega) = & -\frac{4g^2}{\omega + i\gamma} \left\{ \left[\alpha^{(0)(1)} * \text{Im} \left[\alpha^{(1)(0)} * \rho_{12}^{(1)(0)} \right] \right]_\tau (\omega) + \left[\alpha^{(1)(0)} * \text{Im} \left[\alpha^{(0)(1)} * \rho_{12}^{(1)(0)} \right] \right]_\tau (\omega) \right. \\ & \left. + \left[\alpha^{(1)(0)} * \text{Im} \left[\alpha^{(1)(0)} * \rho_{12}^{(0)(1)} \right] \right]_\tau (\omega) \right\}, \end{aligned} \quad (\text{S.18})$$

in terms of the first-order quantities in the pump field

$$\alpha^{(1)(0)}(\omega) = \frac{-\tilde{f}_p(\omega - \omega_p)}{\left[\frac{\kappa}{2} + i(\omega_c - \omega)\right] + \frac{g^2\mathcal{N}}{\frac{\gamma + \gamma_\phi}{2} + i(\omega_0 - \omega)}}, \quad \rho_{21}^{(1)(0)}(\omega) = \frac{-ig\alpha^{(1)(0)}(\omega)}{i(\omega_0 - \omega) + \frac{\gamma + \gamma_\phi}{2}}, \quad (\text{S.19})$$

and similarly for the terms linear in the probe. The Fourier transform of the Gaussian pulse shapes is given by

$$\tilde{f}_j(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt e^{-\frac{(t - \tau_j)^2}{2\tau_w^2}} e^{i\omega t} = \tau_w e^{-\omega^2 \tau_w^2 / 2} e^{i\omega \tau_j}, \quad (\text{S.20})$$

where the pulse frequency and arrival time lead to displacement and an additional complex phase in frequency space, respectively. A comparison between the Fourier transform result and the numerical result obtained by solving the differential equations in time domain is shown in Fig. S1.

S6. FORMAL INTEGRATION OF PUMP-PROBE RESPONSE

From Eq. (S.8), the integrated solution for the vectorized density matrix second order in the pump and first order in the probe, $\bar{\rho}^{(2)(1)}$, is obtained as [denoting the free molecular propagator, i.e., the Green's function by

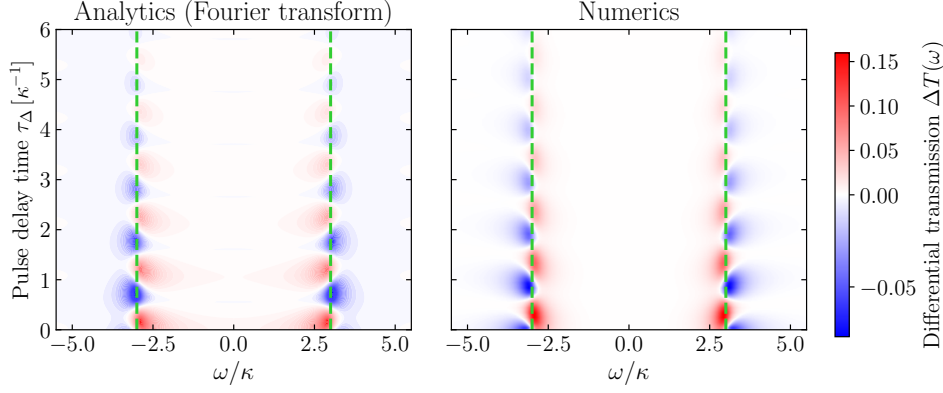


Figure S1: Comparison between analytical (left, Fourier transform) and numerical (right, obtained by solving the differential equations) results for DT spectrum of a 2LS, plotted in a frame rotating at the central pulse frequency. The parameters are $g\sqrt{N} = 3\kappa$, $\tau_w = 0.05\kappa^{-1}$, $\omega_p = \omega_{p'} = \omega_0$, and $\gamma_\phi, \gamma = 0$.

$$\theta(t - t')e^{-i\mathcal{L}_0(t-t')} \equiv \mathcal{G}(t - t')$$

$$\begin{aligned} \bar{\rho}^{(2)(1)}(t) = & -i \int_0^t dt_1 \mathcal{G}(t - t_1) \times \\ & \times \left[\mathcal{L}_{\text{int}}^{(2)(1)}(t_1) \bar{\rho}^{(0)(0)}(t_1) + \mathcal{L}_{\text{int}}^{(2)(0)}(t_1) \bar{\rho}^{(0)(1)}(t_1) + \mathcal{L}_{\text{int}}^{(1)(1)}(t_1) \bar{\rho}^{(1)(0)}(t_1) + \mathcal{L}_{\text{int}}^{(1)(0)}(t_1) \bar{\rho}^{(1)(1)}(t_1) + \mathcal{L}_{\text{int}}^{(0)(1)}(t_1) \bar{\rho}^{(2)(0)}(t_1) \right]. \end{aligned} \quad (\text{S.21})$$

Writing out all terms explicitly in terms of the molecular initial state $\bar{\rho}(0)$, yields the following contributions:

$$\begin{aligned} \bar{\rho}^{(2)(1)}(t) = & -i \int_0^t dt_1 \mathcal{G}(t - t_1) \mathcal{L}_{\text{int}}^{(2)(1)}(t_1) \bar{\rho}(0) \\ & (-i)^2 \int_0^t dt_2 \int_0^{t_2} dt_1 \mathcal{G}(t - t_2) \mathcal{L}_{\text{int}}^{(1)(1)}(t_2) \mathcal{G}(t_2 - t_1) \mathcal{L}_{\text{int}}^{(1)(0)}(t_1) \bar{\rho}(0) \\ & (-i)^2 \int_0^t dt_2 \int_0^{t_2} dt_1 \mathcal{G}(t - t_2) \mathcal{L}_{\text{int}}^{(2)(0)}(t_2) \mathcal{G}(t_2 - t_1) \mathcal{L}_{\text{int}}^{(0)(1)}(t_1) \bar{\rho}(0) \\ & (-i)^2 \int_0^t dt_2 \int_0^{t_2} dt_1 \mathcal{G}(t - t_2) \mathcal{L}_{\text{int}}^{(0)(1)}(t_2) \mathcal{G}(t_2 - t_1) \mathcal{L}_{\text{int}}^{(2)(0)}(t_1) \bar{\rho}(0) \\ & (-i)^2 \int_0^t dt_2 \int_0^{t_2} dt_1 \mathcal{G}(t - t_2) \mathcal{L}_{\text{int}}^{(1)(0)}(t_2) \mathcal{G}(t_2 - t_1) \mathcal{L}_{\text{int}}^{(1)(1)}(t_1) \bar{\rho}(0) \\ & (-i)^3 \int_0^t dt_3 \int_0^{t_3} dt_2 \int_0^{t_2} dt_1 \mathcal{G}(t - t_3) \mathcal{L}_{\text{int}}^{(1)(0)}(t_3) \mathcal{G}(t_3 - t_2) \mathcal{L}_{\text{int}}^{(1)(0)}(t_2) \mathcal{G}(t_2 - t_1) \mathcal{L}_{\text{int}}^{(0)(1)}(t_1) \bar{\rho}(0) \\ & (-i)^3 \int_0^t dt_3 \int_0^{t_3} dt_2 \int_0^{t_2} dt_1 \mathcal{G}(t - t_3) \mathcal{L}_{\text{int}}^{(1)(0)}(t_3) \mathcal{G}(t_3 - t_2) \mathcal{L}_{\text{int}}^{(0)(1)}(t_2) \mathcal{G}(t_2 - t_1) \mathcal{L}_{\text{int}}^{(1)(0)}(t_1) \bar{\rho}(0) \\ & (-i)^3 \int_0^t dt_3 \int_0^{t_3} dt_2 \int_0^{t_2} dt_1 \mathcal{G}(t - t_3) \mathcal{L}_{\text{int}}^{(0)(1)}(t_3) \mathcal{G}(t_3 - t_2) \mathcal{L}_{\text{int}}^{(1)(0)}(t_2) \mathcal{G}(t_2 - t_1) \mathcal{L}_{\text{int}}^{(1)(0)}(t_1) \bar{\rho}(0). \end{aligned} \quad (\text{S.22})$$

The last term corresponds to the result that would be obtained for conventional free-space spectroscopy with well-separated pulses, i.e., two linear applications of the pump field followed by application of the probe field [6].

S7. NARROWBAND SPECTROSCOPY OF POLARITONS

The results presented in the main text were obtained by broadband excitation of the polaritons, i.e., by exciting an (equal) superposition of upper and lower polariton. Instead, one may also consider narrowband excitation of the polaritons, by choosing a pulse width in frequency domain $\tau_w^{-1} < g\sqrt{N}$ and fitting the pulse carrier frequency to the

polariton frequencies. Figs. S2(b), (c) show the result obtained for narrowband pump-probe excitation of the lower polariton. In this case, no Rabi oscillations are observed since no superposition of upper and lower polariton is excited. Narrowband excitation of the upper polariton shows a similar result.

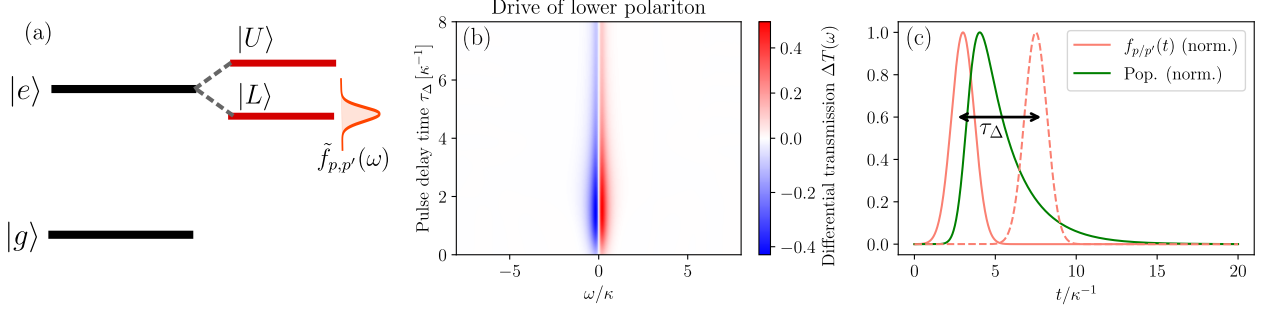


Figure S2: (a) The envelope of the pump and probe pulses in frequency domain is fitted to the lower polariton frequency. (b) DT spectrum for excitation of the lower polariton. We chose $g\sqrt{N} = 3\kappa$ and a pulse width of $\tau_w = \kappa^{-1}$. Note that, since we are in a rotating frame, $\omega = 0$ corresponds to the central pulse (and therefore the lower polariton) frequency. (c) Time evolution of molecular population created by the pump.

S8. EFFECTS OF DISORDER ON DT

To study the effect of inhomogeneous broadening, we employ a simplified model in which the molecular ensemble is divided into two (equal) subensembles with shifted transition frequencies, $\omega_{\pm} = \omega_0 \pm \delta$. In Fig. S3, we illustrate the effect of the degree of disorder δ on the DT spectra. Generally, we find that the DT appears quite robust with increasing disorder. At larger degrees of disorder, the dark states are appearing in the spectrum at $\omega = 0$, and there is a slight increase in the vacuum Rabi splitting, consistent with previous theoretical treatments [7–9]. Furthermore, the bright state lifetime [as extracted from the DQC (2, −1) contribution] decays faster with increasing disorder.

S9. BRIGHT AND DARK STATES IN MEAN-FIELD THEORY

We show here how molecular dark and bright states are arising in the mean-field approach (for a 2LS) [10]. The total population is given by

$$p_{\text{tot}} = \sum_{j=1}^{\mathcal{N}} \langle \sigma_j^\dagger \sigma_j \rangle, \quad (\text{S.23})$$

which in mean field theory becomes

$$p_{\text{tot}} = \mathcal{N} \langle \sigma^\dagger \sigma \rangle. \quad (\text{S.24})$$

Using that the general population of the k -th collective mode is given by $p_k = \langle \sigma_k^\dagger \sigma_k \rangle$, the population of the bright mode (corresponding to $k = 0$) is given by

$$p_B = p_{k=0} = \frac{1}{\mathcal{N}} \sum_{j,j'=1}^{\mathcal{N}} \langle \sigma_j^\dagger \sigma_{j'} \rangle = \frac{1}{\mathcal{N}} \sum_{j=1}^{\mathcal{N}} \langle \sigma_j^\dagger \sigma_j \rangle + \frac{1}{\mathcal{N}} \sum_{j \neq j'} \langle \sigma_j \rangle^* \langle \sigma_{j'} \rangle, \quad (\text{S.25})$$

which in mean field theory becomes

$$p_B = \langle \sigma^\dagger \sigma \rangle + (\mathcal{N} - 1) |\langle \sigma \rangle|^2 \quad (\text{S.26})$$

The dark state population $p_D = p_{\text{tot}} - p_B$ is therefore given

$$p_D = (\mathcal{N} - 1) \langle \sigma^\dagger \sigma \rangle - (\mathcal{N} - 1) |\langle \sigma \rangle|^2. \quad (\text{S.27})$$

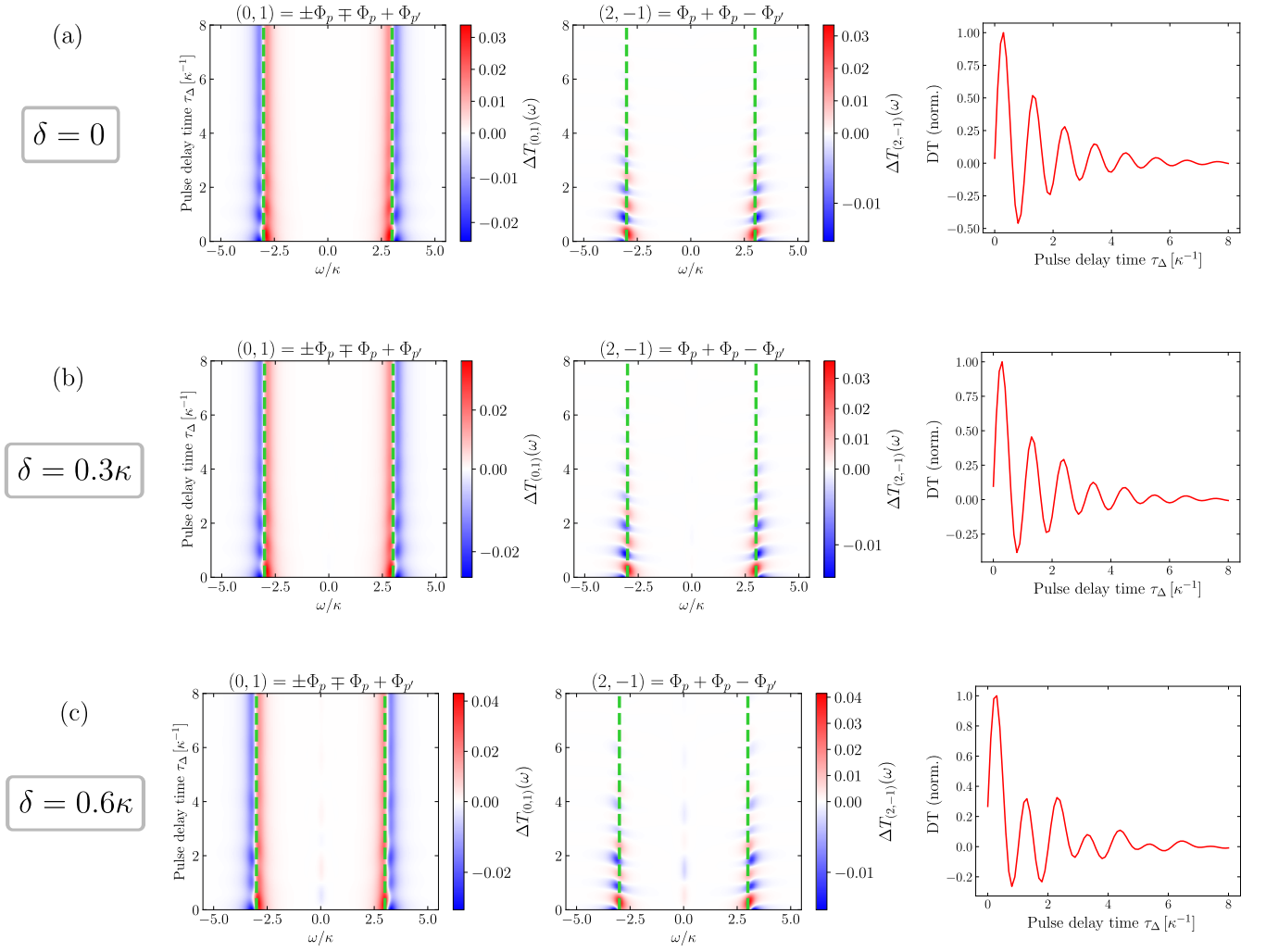


Figure S3: DT for varying degrees of disorder (a) $\delta = 0$, (b) $\delta = 0.3\kappa$, and (c) $\delta = 0.6\kappa$. Left and middle columns show different phase contributions, while the right column presents a cut through the $(2, -1)$ contribution at $\omega = g\sqrt{\mathcal{N}}$ (normalized for comparison across different cases). We chose $g\sqrt{\mathcal{N}} = 3\kappa$, $\gamma_\phi = 0.2\kappa$.

In the large $\mathcal{N}$ limit, this can be expressed as

$$p_B \approx \langle \sigma^\dagger \sigma \rangle + \mathcal{N} |\langle \sigma \rangle|^2, \quad (\text{S.28a})$$

$$p_D \approx \mathcal{N} (\langle \sigma^\dagger \sigma \rangle - |\langle \sigma \rangle|^2). \quad (\text{S.28b})$$

Therefore, the bright state corresponds to molecular coherence in the large $\mathcal{N}$ limit.

S10. DETAILS ON PHASE EXPANSION

For an arbitrary multilevel system described by a Hamiltonian $\mathcal{H}_0$ and a dipole operator $\hat{\mu} = \hat{\mu}^{(+)} + \hat{\mu}^{(-)}$, the mean-field Hamiltonian in the rotating wave approximation is given by

$$\mathcal{H}_{\text{MF}} = \mathcal{H}_0 + E_0 \left(\alpha^*(t) \hat{\mu}^{(-)} + \alpha(t) \hat{\mu}^{(+)} \right). \quad (\text{S.29})$$

We expand the cavity field and density matrix in terms of the input fields' amplitudes and phases as

$$\alpha(t) = \sum_{n,m,\vec{v}} \eta_p^n \eta_{p'}^m e^{-i\vec{v}\cdot\vec{\Phi}} \alpha_{\vec{v}}^{(n)(m)}(t), \quad \rho(t) = \sum_{n,m,\vec{v}} \eta_p^n \eta_{p'}^m e^{-i\vec{v}\cdot\vec{\Phi}} \rho_{\vec{v}}^{(n)(m)}(t), \quad (\text{S.30})$$

where $\vec{\Phi} = (\Phi_p, \Phi_{p'})^\top$ denotes the vector of pump and probe phases and $\vec{v} = (v_p, v_{p'})^\top$ denotes the coefficients of the phases, respectively. Possible values for the phase coefficients in $(n), (m)$ -th order are $v_p(n) = \sum_{j=1}^n s_j$, $v_{p'}(m) = \sum_{j=1}^m s_j$, where each $s_j = \pm 1$. Note that complex conjugation corresponds to flipping the sign of the phase, i.e., $[\alpha_{\vec{v}}^{(n)(m)}]^* = \alpha_{-\vec{v}}^{(n)(m)*}$.

The equation for the $(n)(m)$ -th order cavity field including phases on the pump and probe input fields reads

$$\dot{\alpha}^{(n)(m)}(t) = -\left(\frac{\kappa}{2} + i\omega_c\right) \alpha^{(n)(m)} - iE_0 \mathcal{N} \text{Tr}[\hat{\mu}^{(-)} \rho^{(n)(m)}] - \delta_{1n} \delta_{0m} f_p(t - \tau_p) e^{-i\omega_p t} e^{-i\Phi_p} - \delta_{0n} \delta_{1m} f_{p'}(t - \tau_{p'}) e^{-i\omega_{p'} t} e^{-i\Phi_{p'}}. \quad (\text{S.31})$$

It is now straightforward to see that, at first order, the only phase contributions of the cavity field which are driven by the input pulses are $(1, 0)$ for the field created by the pump and $(0, 1)$ for the field created by the probe

$$\dot{\alpha}_{(1,0)}^{(1)(0)} = -\left(\frac{\kappa}{2} + i\omega_c\right) \alpha_{(1,0)}^{(1)(0)} - iE_0 \mathcal{N} \text{Tr}[\hat{\mu}^{(-)} \rho_{(1,0)}^{(1)(0)}] - f_p(t - \tau_p) e^{-i\omega_p t}, \quad (\text{S.32a})$$

$$\dot{\alpha}_{(0,1)}^{(0)(1)} = -\left(\frac{\kappa}{2} + i\omega_c\right) \alpha_{(0,1)}^{(0)(1)} - iE_0 \mathcal{N} \text{Tr}[\hat{\mu}^{(-)} \rho_{(0,1)}^{(0)(1)}] - f_{p'}(t - \tau_{p'}) e^{-i\omega_{p'} t}, \quad (\text{S.32b})$$

while the other possible phase combinations at first order $(-1, 0)$ and $(0, -1)$ are contained in the complex conjugate cavity fields. All other phase components are not driven at first order and are therefore zero. The positive-phase cavity fields can create positive-phase density matrices in first order via the application of $\hat{\mu}^{(+)}$ to a ket (creation of excitation) or to a bra (annihilation of excitation):

$$\dot{\rho}_{(1,0)}^{(1)(0)} = -i[\mathcal{H}_0, \rho_{(1,0)}^{(1)(0)}] + \mathcal{D}[\rho_{(1,0)}^{(1)(0)}] - iE_0 \alpha_{(1,0)}^{(1)(0)} [\hat{\mu}^{(+)}, \rho(0)], \quad (\text{S.33a})$$

$$\dot{\rho}_{(0,1)}^{(0)(1)} = -i[\mathcal{H}_0, \rho_{(0,1)}^{(0)(1)}] + \mathcal{D}[\rho_{(0,1)}^{(0)(1)}] - iE_0 \alpha_{(0,1)}^{(0)(1)} [\hat{\mu}^{(+)}, \rho(0)], \quad (\text{S.33b})$$

while the complex conjugate processes correspond to the application of $\hat{\mu}^{(-)}$, thereby imprinting negative phase contributions onto the density matrix.

Beyond first order, the cavity field is not driven directly anymore. Instead, the density matrix is now driven by lower orders of α , which determines the possible phase contributions in higher orders. The phase combinations for the $(n)(m)$ -th order density matrix driven by lower orders of cavity field and density matrix are given by

$$\dot{\rho}_{\vec{v}}^{(n)(m)} = -i[\mathcal{H}_0, \rho_{\vec{v}}^{(n)(m)}] + \mathcal{D}[\rho_{\vec{v}}^{(n)(m)}] - iE_0 \sum_{j=0}^{n-1} \sum_{j'=0}^{m-1} \sum_{\vec{u}+\vec{w}=\vec{v}} \left(\alpha_{\vec{u}}^{(n-j)(m-j')} [\hat{\mu}^{(+)}, \rho_{\vec{w}}^{(j)(j')}] + \alpha_{\vec{u}}^{(n-j)(m-j')*} [\hat{\mu}^{(-)}, \rho_{\vec{w}}^{(j)(j')}] \right), \quad (\text{S.34})$$

where the possible values for $\vec{u}, \vec{w}$ are determined by the phase contributions in lower orders. This drives the phase contributions for the cavity field at nonlinear orders $n + m > 1$ as

$$\dot{\alpha}_{\vec{v}}^{(n)(m)}(t) = -\left(\frac{\kappa}{2} + i\omega_c\right) \alpha_{\vec{v}}^{(n)(m)} - iE_0 \mathcal{N} \text{Tr}[\hat{\mu}^{(-)} \rho_{\vec{v}}^{(n)(m)}]. \quad (\text{S.35})$$

S11. EXPLICIT PHASE EXPANSION FOR THREE-LEVEL SYSTEMS

Let us explicitly consider the phase expansion for a three-level system (3LS), which can, e.g., serve as a simplified model for an anharmonic vibrational mode, described by the dipole operator

$$\hat{\mu} = \mu_{12} \hat{\sigma}_{12} + \mu_{23} \hat{\sigma}_{23} + \text{H.c.}, \quad (\text{S.36})$$

where $\hat{\sigma}_{12} = |1\rangle\langle 2|$ and $\hat{\sigma}_{23} = |2\rangle\langle 3|$ are the corresponding lowering operators. The mean-field Hamiltonian of a 3LS then reads

$$\mathcal{H}_{\text{MF}} = \omega_2 \hat{\sigma}_{12}^\dagger \hat{\sigma}_{12} + \omega_3 \hat{\sigma}_{23}^\dagger \hat{\sigma}_{23} + g_{12} \left(\hat{\sigma}_{12}^\dagger \alpha(t) + \text{h.c.} \right) + g_{23} \left(\hat{\sigma}_{23}^\dagger \alpha(t) + \text{h.c.} \right), \quad (\text{S.37})$$

with the transition frequencies ω_2, ω_3 (we set $\omega_1 = 0$), and $g_{12} = \mu_{12}E_0$, $g_{23} = \mu_{23}E_0$ describe the coupling strengths of the $|1\rangle\langle 2|$ and $|2\rangle\langle 3|$ transitions to the cavity field, respectively. The 2LS limit can simply be obtained by setting $g_{23} = 0$. In addition, we will consider dephasing affecting the transitions which can be described by the dissipator

$$\mathcal{D}[\rho] = \gamma_\phi \left[\hat{\sigma}_{12}^\dagger \hat{\sigma}_{12} \rho \hat{\sigma}_{12}^\dagger \hat{\sigma}_{12} - \frac{1}{2} \left\{ \rho, \hat{\sigma}_{12}^\dagger \hat{\sigma}_{12} \right\} \right] + \gamma_\phi \left[\hat{\sigma}_{23}^\dagger \hat{\sigma}_{23} \rho \hat{\sigma}_{23}^\dagger \hat{\sigma}_{23} - \frac{1}{2} \left\{ \rho, \hat{\sigma}_{23}^\dagger \hat{\sigma}_{23} \right\} \right], \quad (\text{S.38})$$

where $\{\cdot, \cdot\}$ denotes the anticommutator.

a. First order. Assuming identical frequencies for pump and probe $\omega_p = \omega_{p'}$ and an initial state $\rho(0) = |g\rangle\langle g|$, the equations for the first order coherences in a frame rotating at ω_p are given by

$$\dot{\rho}_{21,(1,0)}^{(1)(0)} = - \left(\frac{\gamma_\phi}{2} - i\Delta_{12} \right) \rho_{21,(1,0)}^{(1)(0)} - ig_{12} \alpha_{(1,0)}^{(1)(0)}, \quad (\text{S.39a})$$

$$\dot{\rho}_{21,(0,1)}^{(0)(1)} = - \left(\frac{\gamma_\phi}{2} - i\Delta_{12} \right) \rho_{21,(0,1)}^{(0)(1)} - ig_{12} \alpha_{(0,1)}^{(0)(1)}, \quad (\text{S.39b})$$

while the equation for the first-order fields are given by

$$\dot{\alpha}_{(1,0)}^{(1)(0)} = - \left(\frac{\kappa}{2} - i\Delta_c \right) \alpha_{(1,0)}^{(1)(0)} - i\mathcal{N}g_{12}\rho_{21,(1,0)}^{(1)(0)} - f_p(t - \tau_p), \quad (\text{S.40a})$$

$$\dot{\alpha}_{(0,1)}^{(0)(1)} = - \left(\frac{\kappa}{2} - i\Delta_c \right) \alpha_{(0,1)}^{(0)(1)} - i\mathcal{N}g_{12}\rho_{21,(0,1)}^{(0)(1)} - f_{p'}(t - \tau_{p'}), \quad (\text{S.40b})$$

where we define the detunings w.r.t. the central input pulse frequency $\Delta_c = \omega_p - \omega_c$, $\Delta_{12} = \omega_p - \omega_2$, and $\Delta_{23} = \omega_p - (\omega_3 - \omega_2)$. The other phase combinations $(-1, 0)$ and $(0, -1)$ are contained in the complex conjugate cavity fields.

b. Second order. The second order populations and coherences are given by

$$\dot{\rho}_{22,(0,0)}^{(2)(0)} = -ig_{12}\alpha_{(1,0)}^{(1)(0)}\rho_{12,(-1,0)}^{(1)(0)} + ig_{12}\alpha_{(-1,0)}^{(1)(0)*}\rho_{21,(1,0)}^{(1)(0)}, \quad (\text{S.41a})$$

$$\dot{\rho}_{22,(1,-1)}^{(1)(1)} = -ig_{12}\alpha_{(1,0)}^{(1)(0)}\rho_{12,(0,-1)}^{(0)(1)} + ig_{12}\alpha_{(0,-1)}^{(0)(1)*}\rho_{21,(1,0)}^{(1)(0)}, \quad (\text{S.41b})$$

$$\dot{\rho}_{31,(2,0)}^{(2)(0)} = - \left[\frac{\gamma_\phi}{2} - i(\Delta_{12} + \Delta_{23}) \right] \rho_{31,(2,0)}^{(2)(0)} - ig_{23}\alpha_{(1,0)}^{(1)(0)}\rho_{21,(1,0)}^{(1)(0)}, \quad (\text{S.41c})$$

$$\dot{\rho}_{31,(1,1)}^{(1)(1)} = - \left[\frac{\gamma_\phi}{2} - i(\Delta_{12} + \Delta_{23}) \right] \rho_{31,(1,1)}^{(1)(1)} - ig_{23}\alpha_{(0,1)}^{(0)(1)}\rho_{21,(1,0)}^{(1)(0)} - ig_{23}\alpha_{(1,0)}^{(1)(0)}\rho_{21,(0,1)}^{(0)(1)}, \quad (\text{S.41d})$$

while all second-order cavity fields are not driven and therefore remain zero at all times. The other phase combinations in second order $(-1, 1)$, $(-2, 0)$, and $(-1, -1)$ are obtained as the complex conjugates of the above equations.

c. Third order. The third-order coherences are given by

$$\dot{\rho}_{21,(0,1)}^{(2)(1)} = - \left(\frac{\gamma_\phi}{2} - i\Delta_{12} \right) \rho_{21,(0,1)}^{(2)(1)} - ig_{12}\alpha_{(0,1)}^{(2)(1)} + 2ig_{12}\alpha_{(0,1)}^{(0)(1)}\rho_{22,(0,0)}^{(2)(0)} + 2ig_{12}\alpha_{(1,0)}^{(1)(0)}\rho_{22,(-1,1)}^{(1)(1)} - ig_{23}\alpha_{(-1,0)}^{(1)(0)*}\rho_{31,(1,1)}^{(1)(1)}, \quad (\text{S.42a})$$

$$\dot{\rho}_{21,(2,-1)}^{(2)(1)} = - \left(\frac{\gamma_\phi}{2} - i\Delta_{12} \right) \rho_{21,(2,-1)}^{(2)(1)} - ig_{12}\alpha_{(2,-1)}^{(2)(1)} + 2ig_{12}\alpha_{(1,0)}^{(1)(0)}\rho_{22,(1,-1)}^{(1)(1)} - ig_{23}\alpha_{(0,-1)}^{(0)(1)*}\rho_{31,(2,0)}^{(2)(0)}, \quad (\text{S.42b})$$

$$\dot{\rho}_{32,(0,1)}^{(2)(1)} = - (\gamma_\phi - i\Delta_{23}) \rho_{32,(0,1)}^{(2)(1)} - ig_{23}\alpha_{(0,1)}^{(0)(1)}\rho_{22,(0,0)}^{(2)(0)} - ig_{23}\alpha_{(1,0)}^{(1)(0)}\rho_{22,(-1,1)}^{(1)(1)} + ig_{12}\alpha_{(-1,0)}^{(1)(0)*}\rho_{31,(1,1)}^{(1)(1)}, \quad (\text{S.42c})$$

$$\dot{\rho}_{32,(2,-1)}^{(2)(1)} = - (\gamma_\phi - i\Delta_{23}) \rho_{32,(2,-1)}^{(2)(1)} - ig_{23}\alpha_{(1,0)}^{(1)(0)}\rho_{22,(1,-1)}^{(1)(1)} + ig_{12}\alpha_{(0,-1)}^{(0)(1)*}\rho_{31,(2,0)}^{(2)(0)}. \quad (\text{S.42d})$$

The third-order equations for the fields are given by

$$\dot{\alpha}_{(0,1)}^{(2)(1)} = - \left(\frac{\kappa}{2} - i\Delta_c \right) \alpha_{(0,1)}^{(2)(1)} - i\mathcal{N}g_{12}\rho_{21,(0,1)}^{(2)(1)} - i\mathcal{N}g_{23}\rho_{32,(0,1)}^{(2)(1)}, \quad (\text{S.43a})$$

$$\dot{\alpha}_{(2,-1)}^{(2)(1)} = - \left(\frac{\kappa}{2} - i\Delta_c \right) \alpha_{(2,-1)}^{(2)(1)} - i\mathcal{N}g_{12}\rho_{21,(2,-1)}^{(2)(1)} - i\mathcal{N}g_{23}\rho_{32,(2,-1)}^{(2)(1)}. \quad (\text{S.43b})$$

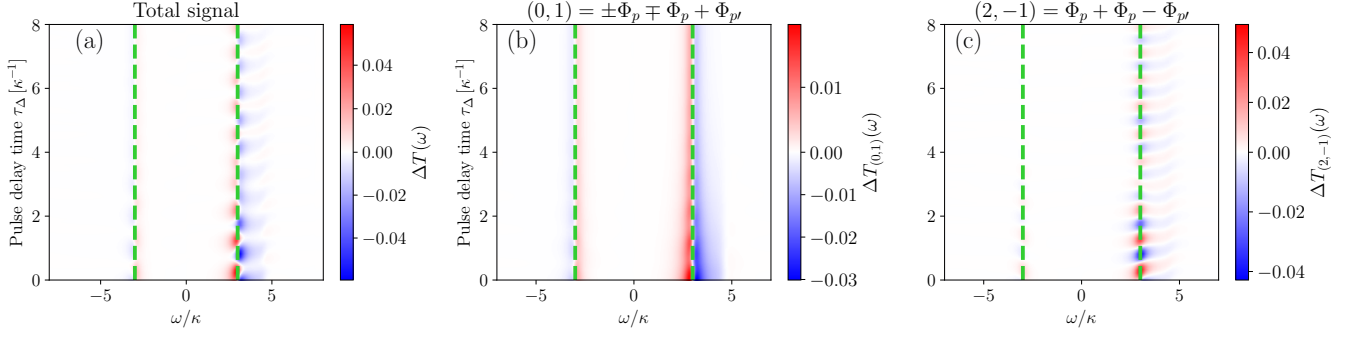


Figure S4: (a) Total DT signal as a function of pulse delay time for a 3LS, as well as phase contributions (b) $(0,1)$, and (c) $(2,-1)$. We chose $g_{12}\sqrt{\mathcal{N}} = 3\kappa$, $g_{23} = \sqrt{2}g_{12}$, $\omega_c = \omega_p = \omega_{p'} = \omega_2$, and $\omega_3 - \omega_2 = 1.8\omega_2$. The dephasing was set to $\gamma_\phi = 0.1\kappa$.

The other phase combinations $(0,-1)$, and $(-2,1)$ are described by the complex conjugate cavity fields. The resulting DT spectrum for the different phase contributions is plotted in Fig. S4, showing that for a 3LS the $(2,-1)$ -contribution also exists for pulse delay times $> (\kappa + \gamma_\phi)^{-1}$.

S12. EXTENSION TO MULTIMODE CAVITIES

For many scenarios and applications (e.g., studies of polariton transport, polariton condensation,...), the picture of a single cavity mode does not suffice and instead many photonic modes inside the cavity are required. We sketch here how the mean-field treatment can be extended in a straightforward way to also account for such more complicated electromagnetic environments, namely by grouping the molecular system into subsystems [11].

To this end, we consider a planar cavity which extends a width W perpendicular (transverse) to the cavity axis. To enable a tractable numerical treatment, one can coarse-grain the transverse extent into N_k discrete points at distance Δr , $W = N_k \Delta r$. Assuming the total number of molecules $\mathcal{N}$ inside the cavity to be uniformly distributed along the transverse direction, every one of the grid points is assigned $N_E = \mathcal{N}/N_k$ molecules interacting with N_k photon modes. This is described by the Hamiltonian

$$\mathcal{H} = \sum_{k=1}^{N_k} \omega_k \hat{a}_k^\dagger \hat{a}_k + \sum_{n=1}^{N_k} \sum_{j=1}^{N_E} \left[\mathcal{H}_0^{nj} + E_0 \sum_{k=1}^{N_k} \left(\hat{a}_k \hat{\mu}_{nj}^{(+)} e^{ikr_n} + \text{h.c.} \right) \right], \quad (\text{S.44})$$

where the index $n = 1, \dots, N_k$ labels the position while $j = 1, \dots, N_E$ labels the particular molecule at a certain position. For the cavity, one may assume a quadratic dispersion $\omega_k = \omega_c + k^2 c^2 / (2\omega_c)$, with ω_c the cavity frequency at $k = 0$. This parabolic approximation is well-known to accurately capture the mode structure of planar cavities near normal incidence. Within the mean-field approximation, the cavity field is assumed to be classical and all molecules at a position r_n are assumed to be identical, i.e., $\mathcal{H}_0^{nj} = \mathcal{H}_0^n$, $\hat{\mu}_{nj} = \hat{\mu}_n$, leading to the simplified Hamiltonian

$$\mathcal{H}_{\text{MF}} = \sum_{n=1}^{N_k} \mathcal{H}_0^n + E_0 \sum_{k=1}^{N_k} \sum_{n=1}^{N_k} \left(\alpha_k(t) \hat{\mu}_n^{(+)} e^{ikr_n} + \text{h.c.} \right). \quad (\text{S.45})$$

The classical evolution of the N_k cavity modes driven by pump and probe input pulses with wavevectors k_p and $k_{p'}$ transverse to the cavity axis is given by

$$\dot{\alpha}_k(t) = - \left(i\omega_k + \frac{\kappa}{2} \right) \alpha_k(t) - iE_0 N_E \sum_{n=1}^{N_k} \left\langle \hat{\mu}_n^{(-)}(t) \right\rangle e^{-ikr_n} - \sum_{j=p,p'} \eta_j \delta_{k_j,k} f_j(t - \tau_j) e^{-i\omega_j t}, \quad (\text{S.46})$$

where only the modes corresponding to $k = k_p, k_{p'}$ are driven by the input pulses and we assume an equal decay rate κ for all cavity modes. This set of equations can now, e.g., be used to model pump-probe spectroscopy with pulses

entering the cavity at oblique angles.

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